

LJUNGBYHED, Sweden

WMO: 020001

Lat: 56.083N

Lon: 13.213E

Elev: 43

StdP: 100.81

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-13.0	-9.7	-15.2	1.0	-11.7	-12.0	1.3	-8.2	13.5	4.9	12.1	4.1	0.9	110	0.615

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	27.9	19.1	26.0	18.6	24.0	17.6	20.6	25.2	19.6	23.8	18.5	22.6	4.2	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.1	13.9	22.6	18.0	13.0	21.3	17.0	12.2	20.2	59.3	25.3	55.7	23.8	52.5	22.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.1	9.6	8.5	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB													
			WB	-17.8	22.3	4.9	1.3	-21.4	23.2	-24.2	24.0	-27.0	24.7	-30.5	25.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.1	0.2	-0.1	2.2	7.0	11.1	14.3	17.5	16.6	12.8	9.0	4.9	1.3
	DBStd	7.10	4.52	3.90	3.87	3.43	3.44	2.92	2.78	2.55	2.72	3.45	3.67	4.91
	HDD10.0	1447	303	282	242	102	27	2	0	0	8	59	154	268
	HDD18.3	3771	562	515	500	340	226	125	51	66	167	290	403	527
	CDD10.0	762	0	0	1	12	60	131	232	205	91	28	2	0
	CDD18.3	43	0	0	0	0	1	4	24	13	1	0	0	0
	CDH23.3	542	0	0	0	2	26	83	314	112	6	0	0	0
	CDH26.7	109	0	0	0	0	1	13	79	16	0	0	0	0
(14) Wind	WSAvg	3.8	4.3	4.2	4.2	3.9	3.7	3.7	3.2	3.2	3.6	3.7	3.9	3.9
(15) Precipitation	PrecAvg	846	72	58	49	41	56	83	85	94	78	82	73	75
	PrecMax	1120	172	151	116	75	117	189	229	199	175	179	149	152
	PrecMin	588	6	9	16	8	24	0	2	23	13	36	23	18
	PrecStd	123	41	27	25	22	25	46	54	48	40	35	29	35
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.0	8.9	14.9	21.9	26.1	28.6	31.0	28.3	24.1	19.0	13.0	10.8
		MCWB	8.9	6.7	9.1	13.8	17.9	18.9	20.2	19.1	18.3	15.5	11.9	9.9
	2%	DB	8.0	7.3	11.1	19.0	23.8	25.8	29.0	26.2	21.1	16.1	11.8	9.1
		MCWB	7.5	6.2	7.7	12.1	16.0	18.1	19.8	18.9	16.9	14.2	10.8	8.6
	5%	DB	7.0	6.3	9.0	16.3	21.1	23.1	27.0	24.1	19.1	14.9	10.8	8.1
		MCWB	6.3	5.4	6.8	10.5	14.5	16.7	19.1	18.2	15.6	13.4	9.7	7.6
	10%	DB	5.9	5.2	7.8	14.0	18.9	20.9	24.8	22.1	17.8	13.9	9.8	7.1
		MCWB	5.2	4.3	6.2	9.4	13.3	15.9	18.2	17.5	14.8	12.5	9.1	6.6
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.1	7.5	10.5	15.0	19.4	20.9	22.6	21.5	18.8	16.1	12.3	10.0
		MCDB	9.6	8.0	12.8	20.4	24.7	26.0	28.4	25.4	22.5	18.3	12.8	10.5
	2%	WB	7.4	6.6	8.3	12.7	17.2	18.8	21.0	19.8	17.4	14.7	10.8	8.6
		MCDB	7.7	7.1	10.1	17.8	21.7	23.5	26.1	23.5	20.4	15.9	11.6	8.9
	5%	WB	6.2	5.5	7.1	11.2	15.5	17.6	20.1	18.8	16.1	13.7	9.8	7.5
		MCDB	6.6	6.1	8.4	14.9	19.6	22.0	24.7	22.4	18.5	14.8	10.4	7.9
	10%	WB	5.1	4.3	5.9	9.9	13.9	16.3	19.1	18.0	15.2	12.8	8.9	6.6
		MCDB	5.6	5.1	7.3	13.2	17.9	20.0	23.3	21.3	17.1	13.8	9.3	6.9
(35) Mean Daily Temperature Range	5% DB	MDBR	4.6	4.9	7.3	10.3	11.1	10.0	11.1	9.5	8.4	6.9	5.0	4.6
		MCDBR	3.9	5.4	9.6	15.1	16.0	14.9	15.7	13.2	11.1	7.8	4.7	4.3
	5% WB	MCWBR	4.0	4.5	6.9	9.1	9.0	8.1	7.6	6.6	7.2	5.9	4.1	4.3
		MCDWR	4.0	4.9	8.3	12.7	13.0	12.9	12.5	10.9	10.1	7.3	4.4	4.3
(40) Clear-Sky Solar Irradiance	taub		0.308	0.332	0.354	0.390	0.387	0.380	0.403	0.403	0.371	0.355	0.334	0.301
		taud	2.256	2.280	2.301	2.270	2.330	2.374	2.332	2.337	2.412	2.386	2.318	2.242
	Ebn at Noon		586	722	803	826	852	863	832	805	782	691	551	494
		Edh at Noon	55	78	97	116	116	112	115	108	87	70	52	44
(44) All-Sky Solar Radiation	RadAvg		0.43	1.02	2.39	4.05	5.26	5.56	5.16	4.15	2.93	1.42	0.54	0.29
	RadStd		0.08	0.16	0.25	0.52	0.51	0.46	0.62	0.33	0.31	0.21	0.10	0.03

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (64 neighbors)	+0.41	N/A	N/A	N/A	N/A	N/A	N/A	-164	N/A	N/A	N/A

Nomenclature: See separate page